

Linear Regression

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1 Generalized Least Squares (GLS)

Largely adapted from the respective Wikipedia article [1].

1.1 Model definition

In Generalized Least Squares (GLS), we define an outcome's mean to be a linear function of a set of predictors:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

Where:

- $\mathbf{y} \in \mathbb{R}^N$: outcome vector.
 - N : number of samples
- $\mathbf{X} \in \mathbb{R}^{N \times K}$: design matrix.
 - K : number of predictors
- $\boldsymbol{\beta} \in \mathbb{R}^K$: coefficients vector.
- $\boldsymbol{\epsilon} \in \mathbb{R}^N$: error term.

Because of our conditional mean definition earlier, the error term, $\boldsymbol{\epsilon}$, has a mean of zero for a given set of predictor values ($\mathbf{X}$):

$$\mathbb{E}[\boldsymbol{\epsilon}|\mathbf{X}] = \mathbf{0}$$

We also assume that the variance of the error term given $\mathbf{X}$ is described by an invertible covariance matrix, $\boldsymbol{\Omega} \in \mathbb{R}^{N \times N}$:

$$\text{Cov}[\boldsymbol{\epsilon}|\mathbf{X}] = \boldsymbol{\Omega}$$

We further assume that the error term follows a multivariate normal distribution with mean $\mathbf{0}$ and covariance $\boldsymbol{\Omega}$:

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega})$$

And so it follows that:

$$\mathbf{y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Omega})$$

1.2 Least squares estimate of β

We denote a candidate estimate for the β vector as $\mathbf{b}$ and define its residual vector as $\mathbf{y} - \mathbf{Xb}$. The goal of GLS is to find the estimate of β that maximizes the likelihood of the data given the above model, which we can calculate using the probability density function of a multivariate normal distribution [2]:

$$\frac{1}{\sqrt{(2\pi)^K |\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

We can plug in $\Sigma = \Omega$, $\mathbf{x} = \mathbf{y}$, and $\boldsymbol{\mu} = \mathbf{Xb}$. The $\mathbf{b}$ that maximizes this likelihood function will be the same that minimizes the inside of the exponent (without the negative), which is also the squared Mahalanobis length of the residual vector:

$$\hat{\beta} = \arg \min_b (\mathbf{y} - \mathbf{Xb})^\top \Omega^{-1}(\mathbf{y} - \mathbf{Xb})$$

Through matrix algebra, this is equivalent to:

$$\begin{aligned} \hat{\beta} &= \arg \min_b (\mathbf{y} - \mathbf{Xb})^\top \Omega^{-1}(\mathbf{y} - \mathbf{Xb}) \\ &= \arg \min_b \mathbf{y}^\top \Omega^{-1} \mathbf{y} - \mathbf{y}^\top \Omega^{-1} \mathbf{Xb} - (\mathbf{Xb})^\top \Omega^{-1} \mathbf{y} + (\mathbf{Xb})^\top \Omega^{-1} \mathbf{Xb} \\ &= \arg \min_b \mathbf{y}^\top \Omega^{-1} \mathbf{y} + (\mathbf{Xb})^\top \Omega^{-1} (\mathbf{Xb}) - 2(\mathbf{Xb})^\top \Omega^{-1} \mathbf{y} \end{aligned}$$

We can use calculus to solve for the $\mathbf{b}$ that minimizes this expression by taking the partial derivative with respect to $\mathbf{b}$ and solving for 0:

$$\begin{aligned} \mathbf{0} &= \frac{\partial}{\partial \mathbf{b}} [\mathbf{y}^\top \Omega^{-1} \mathbf{y} + (\mathbf{Xb})^\top \Omega^{-1} (\mathbf{Xb}) - 2(\mathbf{Xb})^\top \Omega^{-1} \mathbf{y}] \\ &= 2\mathbf{X}^\top \Omega^{-1} \mathbf{X} \hat{\beta} - 2\mathbf{X}^\top \Omega^{-1} \mathbf{y} \end{aligned}$$

Which yields:

$$\hat{\beta} = (\mathbf{X}^\top \Omega^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \Omega^{-1} \mathbf{y}$$

1.2.1 Specific case of Ordinary Least Squares

Note that in Ordinary Least Squares (OLS), the covariance matrix is an identity matrix, $\Omega = \mathbf{I}$. That is, the residuals are uncorrelated with each other. This simplifies the above equation to:

$$\begin{aligned} \hat{\beta} &= (\mathbf{X}^\top \Omega^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \Omega^{-1} \mathbf{y} \\ &= (\mathbf{X}^\top \mathbf{I} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{I}^{-1} \mathbf{y} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \end{aligned}$$

1.3 Variance of $\hat{\beta}$

To get the variance of the $\hat{\beta}$ estimate, we only need to focus on the variance of $\mathbf{y}$, as all other terms are not random variables. $\text{Var}[\mathbf{y}] = \mathbf{\Omega}$ since ϵ is independent of $\mathbf{X}\beta$, as shown below:

$$\begin{aligned}\text{Var}[\mathbf{y}] &= \text{Var}[\mathbf{X}\beta + \epsilon] \\ &= \text{Var}[\mathbf{X}\beta] + \text{Var}[\epsilon] \\ &= \mathbf{0} + \text{E}[\text{Var}[\epsilon|\mathbf{X}]] + \text{Var}[\text{E}[\epsilon|\mathbf{X}]] \\ &= \text{E}[\mathbf{\Omega}] + \text{Var}[\mathbf{0}] \\ &= \mathbf{\Omega}\end{aligned}$$

Let's define $\mathbf{A} \in \mathbb{R}^{K \times N}$ as $\mathbf{A} = (\mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{\Omega}^{-1}$, such that $\hat{\beta} = \mathbf{A}\mathbf{y}$. Furthermore, note that since $\mathbf{\Omega}$ is a covariance matrix, it (and its inverse) is symmetric: $\mathbf{\Omega} = \mathbf{\Omega}^\top$. The same symmetry property applies to $\mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{X}$. We can then solve:

$$\begin{aligned}\text{Var}[\hat{\beta}] &= \text{Var}[(\mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{y}] \\ &= \text{Var}[\mathbf{A}\mathbf{y}] \\ &= \mathbf{A} \text{Var}[\mathbf{y}] \mathbf{A}^\top \\ &= \mathbf{A} \mathbf{\Omega} \mathbf{A}^\top \\ &= (\mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{\Omega} (\mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{\Omega}^{-1} \\ &= (\mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{\Omega} \mathbf{\Omega}^{-1} \mathbf{X} (\mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{X})^{-1} \\ &= (\mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{X}) (\mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{X})^{-1}\end{aligned}$$

We can define $\mathbf{B} = \mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{X}$, allowing us to simplify the above equation, $\mathbf{B}^{-1} \mathbf{B} \mathbf{B}^{-1} = \mathbf{B}^{-1}$, yielding:

$$\text{Var}[\hat{\beta}] = (\mathbf{X}^\top \mathbf{\Omega}^{-1} \mathbf{X})^{-1}$$

References

- [1] Wikipedia contributors. Generalized least squares, 2026. URL https://en.wikipedia.org/wiki/Generalized_least_squares. Accessed 2026-05-03.
- [2] Wikipedia contributors. Multivariate normal distribution, 2026. URL https://en.wikipedia.org/wiki/Multivariate_normal_distribution. Accessed 2026-05-03.